F78AB – solutions May 18 exam

Q1. Let
i = money rate of return
i' = real rate of return

As inflation is constant, $(1+i) = (1+i') \times 1.02$ [1]

$570 = 15v^{\frac{4}{12}} + 1.04 \times 15v^{\frac{16}{12}} + (1.04)^2 \times 15v^{\frac{28}{12}} + \cdots$ [1]

$= 15(1+i)^{\frac{8}{12}}/1.04(1.04v + 1.04^2v^2 + 1.04^3v^3 + \cdots)$ [1]

$= 15(1+i)^{\frac{8}{12}}/1.04\left(1/(\frac{1+i}{1.04} - 1)\right)$ (assuming *i*>4%) [1]

$\Longrightarrow 38 = (1+i)^{\frac{8}{12}}/\,(i - .04)$ [1]

$i = 6\% \Rightarrow \text{RHS} = 51.9805$
$i = 7\% \Rightarrow \text{RHS} = 34.8713$
$\Rightarrow i = 0.06 + \frac{51.9805-38}{51.9805-34.8713}.01 = .06817$ [1]

$\Rightarrow i' = 1.06817/1.02 - 1 = 4.723\%$ [1]

Q2. (a)
$(1+i_t) \sim \text{Lognormal}(\mu, \sigma^2) \Leftrightarrow \ln(1+i_t) \sim N(\mu, \sigma^2)$
$\Rightarrow \ln(1+i_t)^8 = \sum_{t=1}^{8} \ln(1+i_t) \sim N(8\mu, 8\sigma^2)$ (since i_t's are independent) [1]

i.e. $(1+i_t)^8 \sim \text{Lognormal}(8\mu, 8\sigma^2)$ [1]

$E[1+i_t] = \exp\left(\mu + \frac{\sigma^2}{2}\right) = 1.05$ [0.5]

$Var(1+i_t) = exp(2\mu + \sigma^2)[\exp(\sigma^2) - 1] = 0.09^2$ [0.5]

$\Rightarrow \frac{0.09^2}{1.05^2} = [\exp(\sigma^2) - 1] \Rightarrow \sigma^2 = 0.007320 \qquad (\sigma = 0.08556)$ [1]

$\exp\left(\mu + \frac{0.007320}{2}\right) = 1.05 \Rightarrow \mu = \ln 1.05 - \frac{0.007320}{2} = 0.04513$ [1]

Let S_8 be the accumulation after 8 years. Then

$E(S_8) = exp\left(8\mu + \frac{8\sigma^2}{2}\right) = \exp(8 \times 0.04513 + 4 \times .00732) = 1.4775$
So expected value of investment = £3m $E(S_8) = £4.432m$ [1]

If PV is calculated then $E(A_8) = 0.67682$ and expected discounted value is £2.0305m

(b) Require $P(S_8 > 1.2 \times E[S_8]) = P(S_8 > 1.7730)$

$\Rightarrow P(\ln S_8 > \ln 1.773)$ where $ln\ S_8 \sim N(8\mu, 8\sigma^2) = N(0.3610, 0.05856)$ [1]
$\Rightarrow P\left(N(0,1) > \frac{\ln 1.773 - 0.3610}{\sqrt{0.05856}}\right) = 1 - P(N(0,1) < 0.875)$ [1]

$= 1 - 0.809 = .191$ [1]

Q3.

(a) $i = \left(1 + \frac{.05}{12}\right)^{12} - 1 = 0.051162$ [1]
$a_{\overline{10|}} = \frac{1-v^{10}}{i} = 7.6784$ [1]

(b) $i = \left(1 - \frac{.05}{12}\right)^{-12} - 1 = 0.051381$ [1]
$a_{\overline{10|}} = \frac{1-v^{10}}{i} = 7.6702$ [1]

(c) $a_{\overline{10|}} = \frac{1-e^{-10\delta}}{e^{\delta}-1} = 7.6743$ $\quad (i = e^{\delta} - 1 = 0.051271)$ [1]

Q4

(a) Let $A(s,t)$ be the accumulation factor at time s of a payment at time t

$A(0,5) = exp\left(\int_0^5 0.02dt\right) = exp[.02t]_0^5 = e^{0.1}$ [1]
$A(5,10) = exp\left(\int_5^{10} 0.004t\ dt\right) = exp[.002t^2]_5^{10} = e^{0.002(100-25)} = e^{0.15}$ [1]
$A(10,12) = exp\left(\int_{10}^{12} 0.003t + 0.0001t^2\ dt\right) = exp[.0015t^2 + .0001t^3/3]_{10}^{12} =$
$\exp(.0015(144 - 100) + .0001\frac{1728-1000}{3}) = e^{0.0903}$ [2]
$PV = \frac{1}{A(0,5)A(5,10)A(10,12)} = \exp(-0.1 - .15 - 0.0903) = 0.7116$ [1]

(NB if 0.00001 used as factor then $A(10,12) = e^{0.068427}$ and $PV = 0.8324$.)

(b) Annualised rate $(1+i)^{-12} = 0.7116$ [1]
$\Rightarrow i = 0.7116^{-\frac{1}{12}} - 1 = 0.02876$ [1]

(NB if 0.00001 used as factor then $i = 0.01540$.)

(c) $PV = \int_2^5 \rho(t) exp\left(-\int_0^t 0.02ds\right) dt$ [1]

$= \int_2^5 e^{-0.05t} exp(-0.02t) dt$ [1]

$= \int_2^5 e^{-0.07t} dt = \left[\frac{e^{-0.07t}}{-0.07}\right]_2^5$ [1]

$= \frac{e^{-0.14} - e^{-0.35}}{0.07} = 2.3524$ [1]

Q5

(a) Present value of the liabilities is $PVL = 80{,}000 \times a_{\overline{15}|} + 100{,}00v^{10}$ at 7% [1]

$= 80{,}000 \times 9.1079 + 100{,}000 * 0.50835 = £779{,}467$ [1]

(b) Discounted mean term of liabilities

$$\frac{(1 \times 80{,}000v + 2 \times 80{,}000v^2 + \cdots + 15 \times 80{,}000v^{15}) + 10 \times 100{,}000v^{10}}{PVL}$$

[1]

$$= \frac{80{,}000(Ia)_{\overline{15}|} + 10 \times 100{,}000v^{10}}{PVL}$$

[1]

$$= \frac{80{,}000(\ddot{a}_{\overline{15}|} - 15v^{15})/i + 10 \times 100{,}000v^{10}}{PVL}$$

[1]

$$= \frac{\frac{80{,}000\left(9.1079 * 1.07 - \frac{15}{1.07^{15}}\right)}{0.07} + 1{,}000{,}000\frac{}{1.07}^{10}}{779467}$$

$$= \frac{80{,}000 * (9.7455 - 5.4367)/0.07 + 1{,}000{,}000/1.07^{10}}{779467}$$

$$= \frac{80{,}000 \times 61.5538 + 508349}{779467} = \frac{5432670}{779467} = 6.9697 \text{ years}$$

[1]

(c) Present value of one unit of each security

$PV_A = 0.08a_{\overline{8}|} + v^8 = 1.059714$

$PV_B = 0.03a_{\overline{25}|} + v^{25} = 0.53386$ [1]

Let A and B denote the nominal amount in each respective security. If PVs of assets and liabilities are equal then

$A \times PV_A + B \times PV_B = 1.059714A + 0.53386B = PVL = 779467$ (*) [1]

If discounted mean terms are equal then

$$\frac{A(0.08(Ia)_{\overline{8}|} + 8v^8) + B(0.03(Ia)_{\overline{25}|} + 25v^{25})}{PVL} = 6.9697$$

[1]

$$\Rightarrow A\left(0.08(Ia)_{\overline{8}|} + 8v^8\right) + B\left(0.03(Ia)_{\overline{25}|} + 25v^{25}\right) = 6.9697 \times 779467$$

[1]

$$\Rightarrow A(0.08 \times 24.7602 + 8 \times 0.58201) + B(0.03 \times 112.3301 + 25 \times 0.18425) = 5{,}428{,}170$$

$\Rightarrow 6.63690A + 7.97615B = 5{,}428{,}170$ (**) [1]

Solve (*) & (**)

Multiply (*) by 6.63690/1.059714
$\Rightarrow 6.63690A + 3.34352B = 4{,}881{,}736$
Deduct from (**)
$\Rightarrow 4.63263B = 546533 \Rightarrow B = £117{,}753$ [1]
$\Rightarrow A = \frac{779467 - 11775 * 0.53386}{1.05971} = £676{,}123$ [1]

(d) The third condition for immunisation is the convexity of the assets is greater than that of the liabilities. It appears that the assets payments are more spread out than those of the liabilities, suggesting it is likely that this condition is met. [2]

Q6
(a) The current value of the forward price of the old contract is:
$90 \times 1.03^4 - 4 \times 1.03^{-2} - 5 \times 1.03^{-4}$
[1]

The current value of the forward price of a new contract is:
$140 - 4 \times 1.03^{-2} - 5 \times 1.03^{-4}$ [1]

The current value of the old forward contract is the difference between the two:
$140 - 90 \times 1.03^4 = £38.70$ [1]

(b) The current value of the forward price of the old contract is:
$90 \times 1.02^{-10} \times 1.03^4 = £83.098$ [1]

The current value of the forward price of a new contract is:
$140 \times 1.02^{-6} = 124.316$ [1]

The current value of the old forward contract is the difference between the two:
$124.316 - 83.098 = £41.22$ [1]

Q7
(a)
$$\left(1 + f_{3,2}\right)^2 = \frac{(1 + y_5)^5}{(1 + y_3)^3} = \frac{(1.03)^5}{(1.025)^3} = 1.07650 \Rightarrow f_{3,2} = 3.755\% \text{ per annum}$$
[2]

(b)
Par yield c_4 given by
$c_4\left(v_{y_1} + v_{y_2}^2 + v_{y_3}^3 + v_{y_4}^4\right) + v_{y_4}^4 = 1$ [1]

$\Rightarrow c_4(1.02^{-1} + 1.0225^{-2} + 1.025^{-3} + 1.0275^{-4}) + 1.0275^{-4} = 1$ [1]

$\Rightarrow \ c_4 = \frac{1 - .89716}{3.7626} = 2.733\%$ per annum [1]

(c)
Par yield is the gross redemption yield for a par yield bond.
Par yield is less than the coupons on the 3% bond.
Therefore a lower proportion of the proceeds of the 3% bond is in the redemption value, which is when spot yields are highest.
So the present value of the proceeds on the bond will be higher than the par yield bond.
Therefore the GRY will be lower than the par yield. [2]